Company for now: Intel

Sector: Technology

Industry: Semiconductors

Frontend + Backend

1. Deal Intake + Document Upload

The analyst first enters the basic loan request:

- Borrower/company name
- Facility type
- Requested amount
- Purpose of loan
- Tenor
- Repayment structure
- Collateral/security, if applicable

Then they upload the relevant documents, such as annual reports, 10-Ks, 10-Qs, audited financial statements and other supporting files.

This is important because the AI should not simply analyse *"Company A"* generally. It should analyse whether **Company A can support this particular loan request**.

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2. Extract and Organise the Relevant Information (Prompt Engineering)

The system processes the uploaded documents and the simulated financial database separately.

A. Extract Business Information from Documents

The LLM reads the uploaded documents and extracts relevant qualitative information, including:

- What the company does
- Industry and business model
- Key revenue drivers
- Main products/services
- Geographic exposure
- Main customers or suppliers, where available
- Capital intensity and major investment plans
- Management/ownership information, where relevant

B. Extract Debt & Financing Information from Documents

The LLM also extracts information relevant to the borrower's existing debt obligations and financing structure, such as:

- Existing debt obligations
- Debt maturity profile
- Interest obligations
- Secured vs. unsecured debt
- Existing financing arrangements
- Relevant financial covenants, where available
- Upcoming debt maturities or refinancing requirements

C. Source Tracking

For important information extracted from the uploaded documents, the system retains the **source document and page number** so that the analyst can verify the information later.

For example:

> **Information:** Intel has significant capital expenditure requirements related to semiconductor manufacturing.
> **Source:** Intel FY2025 Annual Report, p. XX

This creates an **audit trail** linking the information used in the credit assessment back to its original source.

The backend assigns a unique Evidence ID to each important piece of extracted information and stores it in a structured JSON file, together with its original source document, page number and supporting text. For financial figures retrieved from the simulated database, the corresponding CSV data reference is recorded instead.

The backend uses PyMuPDF to check whether the supporting passage can be located on the referenced page of the original PDF before storing the evidence record. References that cannot be located are flagged for analyst review.

These evidence records are subsequently reused during credit memo generation in Section 5 and source verification in Section 6, allowing the system to trace information back to its original source.

D. Retrieve Financial Data from Simulated Database

Historical financial figures and pre-calculated credit metrics are **not extracted by the LLM**.

Instead, the prototype uses a **CSV file to simulate a financial database** containing borrower-specific historical financial data.

The backend uses **Python and Pandas** to:

1. Load the CSV file.
2. Identify the relevant borrower.
3. Retrieve the borrower's historical financial data.
4. Organise the data into a structured format for downstream analysis.

The CSV may contain:

- Revenue
- EBITDA
- EBITDA Margin
- Net Income
- Total Debt
- Net Debt
- Cash
- Operating Cash Flow
- Capex
- Free Cash Flow
- Debt/EBITDA
- Debt/Equity
- Current Ratio
- Interest Coverage
- DSCR

For example:

CSV → Pandas → Filter for Intel → Retrieve 3–5 years of financial data → Structured financial dataset

The structured financial data is then passed to **Section 3: Financial Trend Dashboard & AI Analysis**.

Overall Data Flow

Uploaded Documents
↓
LLM Document Extraction
↓
Business Information + Debt/Financing Information + Source References

Simulated Financial Database (CSV)
↓
Python/Pandas

↓
Historical Financial Data + Pre-calculated Credit Metrics

Both inputs → Section 3: Financial Trend Dashboard + AI Analysis

The LLM therefore focuses on **document understanding and financial interpretation**, while **Pandas handles the retrieval and organisation of structured financial data**.

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3. Financial Trend Dashboard & AI Analysis (Prompt Engineering)

Objective:
Provide visual and AI-assisted analysis of the borrower's historical financial performance and repayment capacity. The system uses the structured financial data retrieved in Section 2 to generate financial charts and uses an LLM to identify meaningful financial trends and relationships. A human analyst reviews and validates the AI analysis before it is passed to the risk assessment stage.

Step 1: Use Structured Financial Data

The system uses the borrower-specific historical financial data and pre-calculated credit metrics retrieved from the simulated financial database in Section 2.

3–5 years of historical data may include:

- **Profitability:** Revenue, EBITDA, EBITDA Margin, Net Profit
- **Leverage:** Total Debt, Net Debt, Debt/EBITDA, Debt/Equity
- **Liquidity:** Cash, Current Ratio, Quick Ratio
- **Cash Flow:** Operating Cash Flow, Capex, Free Cash Flow
- **Debt Servicing:** Interest Expense, Interest Coverage, DSCR

The system uses these values directly for dashboard visualisation and AI analysis. The LLM does **not** extract or calculate the financial metrics.

Step 2: Generate Financial Dashboard

The system generates an interactive dashboard containing charts showing:

- **Profitability:** Revenue, EBITDA and margin trends
- **Leverage:** Debt, Net Debt and Debt/EBITDA trends
- **Liquidity:** Cash and liquidity ratio trends
- **Cash Flow:** Operating Cash Flow, Capex and Free Cash Flow
- **Debt Servicing:** DSCR and Interest Coverage trends

The dashboard allows the analyst to view both the latest financial position and historical changes.

Step 3: AI Analysis of Financial Trends

The LLM receives the same underlying historical financial data and credit metrics used to generate the dashboard.

The AI is instructed to analyse:

A. Individual Trends

- Identify significant increases, decreases or stability in key metrics.
- Compare current performance against historical levels.
- Identify sustained improvements or deterioration.

B. Trend Patterns and Divergences

The AI should look across metrics rather than analyse each number independently.

Examples include:

- **Revenue** ↑ **+ EBITDA** ↓ → potential margin compression
- **Revenue** ↑ **+ Operating Cash Flow** ↓ → weaker cash conversion / working-capital pressure
- **Debt** ↑ **+ EBITDA** ↓ → increasing leverage
- **Debt** ↑ **+ DSCR** ↓ → weakening debt-servicing capacity
- **Cash** ↓ **+ Short-term Debt** ↑ → increasing liquidity pressure

The AI should explain the relationship between the metrics using specific figures and historical comparisons.

C. Link Financial Trends to Business Context

Where relevant business information is available from Section 2, the AI should use it to provide context for the financial trends.

For example:

- Margin compression may be discussed in the context of cost pressures or competitive conditions.
- Rising debt may be considered alongside growth investment or capital expenditure.
- Changes in liquidity may be considered alongside the company's working-capital or seasonal characteristics.

The AI should clearly distinguish between **observed financial trends** and **possible explanations** and should not invent information that is not provided.

D. Repayment Capacity

The AI should assess what the overall financial trends indicate about the borrower's historical ability to meet its debt obligations by considering:

- Profitability and earnings generation
- Cash generation
- Leverage
- Liquidity
- Debt-servicing capacity

The analysis should focus on the **financial evidence supporting or weakening repayment capacity**, rather than formally identifying and ranking credit risks. Detailed risk identification and mitigants are handled in Section 4.

Step 4: Present Charts and AI Analysis Together

The dashboard displays the financial charts alongside concise AI-generated commentary.

Revenue & EBITDA Trend

> Revenue has recovered over the past three years, but EBITDA remains below its historical peak. EBITDA margin has declined from 18% to 12%, indicating that the improvement in revenue has not translated proportionately into profitability.

Debt & DSCR Trend

> Total debt increased by 35% while DSCR declined from 1.7x to 1.2x. The combination indicates that debt has increased faster than the borrower's capacity to service its obligations.

This allows the human analyst to see the **underlying financial evidence and AI interpretation together**.

Step 5: Human Analyst Review

The dashboard acts as the first verification checkpoint before proceeding to the credit risk assessment stage.

The backend uses Python to check that financial figures referenced in the AI-generated analysis match the structured financial data retrieved from the simulated database. Where the AI makes numerical comparisons, Python independently verifies calculations such as percentage changes and historical trends.

The human analyst then reviews the underlying figures, charts and AI-generated interpretations to determine whether the conclusions are supported by the financial evidence.

The analyst can:

- Review the underlying figures and charts
- Validate the AI-generated interpretation
- Add comments or corrections
- Flag information requiring further investigation
- Approve the financial analysis

Once approved, the financial analysis is passed to **Section 4: Identify Key Risks + Suggest Mitigants**.

Example AI Prompt

> **You are a financial analysis AI supporting a corporate credit analyst. You will receive historical financial data and pre-calculated credit metrics retrieved from a simulated financial database, together with relevant business context extracted from the borrower's documents.**
>
> Analyse the borrower's historical financial performance and repayment capacity.
>
> 1. Identify significant trends and changes in profitability, leverage, liquidity, cash flow and debt-servicing metrics.
> 2. Compare current performance with historical levels.
> 3. Identify meaningful relationships, patterns and divergences across financial metrics.
> 4. Explain what these trends indicate about the borrower's historical repayment capacity.
> 5. Where business context is provided, use it to explain possible drivers of the observed financial trends.
> 6. Support observations with specific figures and historical comparisons.
> 7. Clearly distinguish observed facts from possible explanations.
> 8. Do not invent figures, information or unsupported explanations.
> 9. Do not calculate metrics that have not been provided.
> 10. Focus on financial performance and repayment capacity. Detailed credit risk identification and mitigants will be addressed in the next stage.
>
> **Output concise, presentation-ready commentary that can be displayed alongside financial charts on a credit analyst dashboard.**

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4. Identify Key Risks + Suggest Mitigants (Prompt Engineering)

Objective:
Move from generic risk lists to a focused, evidence-backed **risk → evidence → mitigant → proposed term** output.

The AI receives the approved financial analysis from Section 3 together with relevant business, debt and financing information from Section 2 and the loan request from Section 1.

The AI should surface only the **most material 3–5 risks**, tie each to **specific data evidence**, and propose **practical mitigants and loan-term adjustments** for the analyst for CONSIDERATION ONLY, not for making automatic decisions!

Step 1: What the AI should produce (Output Structure)

Instruct the AI to output a compact table (or structured bullets) with four columns/fields per risk:

1. **Identified Risk** – concise label (e.g., “High leverage”, “Weak liquidity”).
2. **Evidence** – specific metrics/trends that justify the risk (with numbers and direction).
3. **Possible Mitigant** – actions that reduce probability or impact of the risk.
4. **Proposed Term / Monitoring Condition** – how the mitigant translates into loan structure, covenants, or reporting.

Example layout:

Identified Risk	Evidence	Possible Mitigant	Proposed Term / Condition
High leverage	Debt/EBITDA increased from 2.1x to 3.7x (3 years)	Limit additional debt; strengthen equity base	Maximum net leverage covenant (e.g., ≤3.5x); equity cure clause
Weak liquidity	Current ratio fell from 1.3x to 0.9x; cash/ST debt ↓	Ensure minimum cash buffer	Minimum liquidity covenant; monthly liquidity reporting
Weak cash conversion	Receivables days up from 45 to 78; outpacing sales	Tighten working-capital management	Maximum DSO covenant; receivables aging report monthly

Customer concentration	Top customer = 40% of revenue	Diversify customer base; monitor key account	Concentration reporting; notification if top customer >35%
Refinancing risk	60% of debt matures within 12 months	Stagger maturities; secure refinancing early	Shorter tenor / refinancing condition; mandatory hedging or amortisation

Any proposed covenant thresholds or loan terms should be treated as **illustrative suggestions for analyst consideration**, rather than automatically determined requirements

Step 2: How the AI should select risks (Prompt Logic)

In order to avoid long, generic lists, engineer the prompt to **(1) Prioritise Materiality** and **(2) Tie each Risk to Specific Evidence**

A. Prioritise Materiality

Ask the AI to:

- Rank risks by **impact on repayment capacity** and **likelihood** (high/medium/low).
- Select only the **top 3–5 risks** that are both material and supported by clear evidence.

You can explicitly say:

“Identify up to five material credit risks that could impair repayment. Prioritise risks where there is clear quantitative evidence (ratio deterioration, trend divergence, concentration, maturity profile).”

B. Tie Each Risk to Specific Evidence

Require each risk to reference:

- Specific ratios or metrics (e.g., “Debt/EBITDA from 2.1x to 3.7x”).
- Time horizon (e.g., “over the last three fiscal years”).
- Any relevant business context (e.g., “in a cyclical industry with falling margins”).

This prevents vague statements like “high leverage is a risk” without showing *why* it matters here. Do remember that these should remain **suggestions for the analyst**, not automatic lending decisions.

Step 3: Designing Mitigants and Terms (Suggestions made by the AI)

For each identified risk, the AI should propose **(1) Mitigants/Risk-Reduction Actions**, **(2) Proposed Terms and Conditions aka Loan Structure Translation**

A. Mitigants (Risk-Reduction Actions)

Examples by risk type:

- **High leverage**
  - Limit further debt; require equity injection; restrict distributions.
- **Weak liquidity**
  - Maintain minimum cash; restrict certain outflows; improve cash forecasting.
- **Weak cash conversion / working capital stress**
  - Tighten credit terms; improve collections; monitor inventory and receivables.
- **Customer concentration**
  - Diversify customer base; deepen contracts with key customers; monitor churn.
- **Refinancing risk**
  - Stagger debt maturities; secure committed facilities earlier; consider amortising structures.

B. Proposed Terms / Conditions (Loan-Structure Translation)

Map each mitigant into something concrete a credit committee can use:

- **Covenants**
  - Maximum leverage ratio (e.g., Net Debt/EBITDA ≤ 3.5x).
  - Minimum interest cover or FCCR (e.g., FCCR ≥ 1.2x).
  - Minimum liquidity (e.g., cash + undrawn RCF ≥ X months of opex).
- **Reporting / Monitoring**
  - Monthly receivables aging and DSO report.
  - Quarterly customer concentration report (top 5 customers as % of revenue).
  - Refinancing status update when >25% of debt is due within 12 months.
- **Structural Protections**
  - Shorter tenor or mandatory amortisation.
  - Equity cure rights (limited frequency, cash-only).
  - Restrictions on distributions or related-party outflows as leverage rises.

Crucially, frame these as **suggestions for the analyst**, not automatic lending decisions. The AI's role is to surface options, not to approve or decline.

Example Prompt

You can implement something like this as your system or user prompt:\

You are a credit analyst AI. You will receive:

- (1) a short business overview, and
  (2) a time-series of key financial metrics and ratios.

  Your task:

  - Identify up to **five material credit risks** that could impair the borrower's ability to repay. Prioritise risks with clear quantitative evidence (ratio trends, concentration, maturity profile).
  - For each risk, provide:
    1. **Identified Risk** – a concise label.
    2. **Evidence** – specific metrics/trends and time period that support the risk.
    3. **Possible Mitigant** – practical actions that could reduce the probability or impact of this risk.
    4. **Proposed Term / Condition** – how the mitigant could be reflected in loan structure, covenants, reporting, or monitoring.

Keep your output structured as a compact table or clearly labelled bullets under each risk. These are **suggestions for the analyst**, not automatic lending decisions. Focus on clarity, relevance, and credit impact.

You then plug in your extracted business info and ratio trends as the user message.

5) How This Improves on Generic Risk Lists

- **From generic to specific**: Instead of "high leverage is a risk," you get "Debt/EBITDA rose from 2.1x to 3.7x; consider a max net leverage covenant and equity cure."
- **From description to action**: Each risk is paired with a mitigant and a concrete term/condition the deal team can negotiate.
- **From exhaustive to focused**: By limiting to 3–5 material risks, the output stays readable and decision-oriented for credit committees.

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5. Generate the Credit Memo with Evidence (Prompt Engineering)

Objective:

Generate a structured, evidence-backed credit memorandum using the information and analysis from Sections 1–4. Important statements should include references to their supporting documents or financial data so that analysts can verify them later.

Step 1: Use Information from Previous Sections

The AI receives:

- **Section 1:** Borrower's loan request and facility details.
- **Section 2:** Extracted business information, existing debt obligations and financial data from the simulated database.
- **Section 3:** Analyst-approved financial analysis and repayment-capacity assessment.
- **Section 4:** Identified credit risks, suggested mitigants and proposed loan terms.

The AI should use this information rather than independently re-extracting or generating new financial figures.

Step 2: Generate the Credit Memo

The AI produces a structured credit memorandum containing:

1. Executive Summary

- Borrower and facility requested
- Purpose of financing
- Key strengths and risks

2. Business Overview

- Business model and industry
- Revenue sources
- Management and ownership, where relevant

3. Financial Analysis

- Revenue and profitability
- Cash flow and liquidity
- Leverage and historical financial trends

4. Repayment Capacity

- Cash generation
- DSCR and debt-servicing capacity
- Existing debt obligations

5. Key Credit Risks

- Financial risks
- Business and operational risks
- Industry and external risks

6. Risk Mitigants / Proposed Loan Structure

- Suggested financial covenants
- Collateral and security
- Tenor and repayment structure
- Monitoring requirements

7. Information Gaps

- Missing financial information or documents
- Conflicting information
- Unsupported statements requiring further investigation

Step 3: Include Supporting Evidence

Important factual statements in the credit memo should include references to their original sources.

For document-based information, the references should contain:

- Source document name
- Page number
- Supporting text or excerpt

For financial figures, references should point to the relevant data in the simulated financial database (CSV).

When generating the credit memo, the LLM is provided with the relevant extracted information and its associated Evidence IDs from Section 2.

The AI is instructed to reference these existing Evidence IDs when making important factual statements, rather than generating its own citations.

The backend then links each referenced Evidence ID to its corresponding source document, page number and supporting passage stored in the JSON file. References that do not exist in the evidence records are flagged.

Step 4: Generate Draft Credit Memo

The system generates a draft credit memo containing the financial analysis, identified risks, suggested mitigants and supporting references.

The draft is then passed to Section 6 for final analyst review and source verification.

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6. Analyst Review + Source Verification (Important)

Objective:

Allow the credit analyst to review the AI-generated credit memo and verify whether important statements are supported by the original uploaded documents or financial data. This establishes a simple audit trail to improve transparency and reduce the risk of AI hallucinations.

Step 1: Display Supporting Evidence

For important statements in the generated credit memo, the system displays the original supporting information.

Each reference should contain:

- **Source Document:** Name of the uploaded PDF or financial database.
- **Page Number:** Location of the supporting information within the PDF.
- **Supporting Text:** Relevant excerpt retrieved from the original document.
- **Financial Data:** Relevant figures from the CSV, where applicable.

When an analyst clicks an Evidence ID in the credit memo, the frontend sends the selected reference to the Python backend. The backend retrieves the corresponding source information from the JSON file and uses PyMuPDF to locate the original document and supporting passage.

The supporting text and page number are then displayed alongside the AI-generated statement, allowing the analyst to compare the two and determine whether the information has been interpreted correctly.

If the source cannot be located, the statement is automatically flagged for review.

Step 2: Human Analyst Verification

After viewing the original supporting evidence, the analyst determines whether the AI-generated statement accurately reflects the information contained in the source document.

The analyst can:

- View the original document and page number.
- Compare the AI-generated statement against the supporting text.
- Check whether the information has been interpreted correctly.
- Approve or flag inaccurate or unsupported statements.
- Make corrections where necessary.

Financial figures and calculations already verified in Section 3 do not need to be checked again unless they have been modified in the final memo.

Important: A citation pointing to an existing document and page does not automatically mean the information is accurate. The analyst must verify whether the evidence actually supports the statement.

Step 3: Maintain a Simple Audit Trail

The backend records the original source references and analyst verification results in a structured JSON file. If an approved statement is subsequently edited, its verification status is reset until the analyst reviews and approves the updated statement.

The audit trail follows:

Source Document / CSV → Extracted Information → AI Analysis → Credit Memo → Analyst Verification

For each important statement, the system records:

- Statement and supporting source reference
- Original supporting text or financial data
- Verification status (Approved / Flagged)
- Analyst corrections, where applicable

The audit trail also incorporates the financial analysis verification performed in Section 3, avoiding unnecessary duplication.

Step 4: Finalise the Credit Memo

Once the analyst has reviewed the important statements and addressed identified errors, the system generates the final Credit Memo in PDF format.

The final memo retains the supporting source references for traceability.

The system therefore acts as an AI-assisted credit analysis tool, with the human analyst responsible for verifying the evidence and finalising the memorandum.

Deliverables:

1. Engineering and Designing Prompt for 2-6
2. Frontend (GUI), Backend
3. Prototype Testing